

Outline of Project Proposal

Significance

Eating is a crucial activity in our lives, and the inability to feed oneself has been shown to lead to feelings of incompetence and shame (Hermann, et al.). It is, therefore, essential to develop assistive feeding devices that can restore independence to individuals who cannot feed themselves due to limited upper limb mobility, or otherwise reduce dependence on caregivers. Caregiver availability is declining due to aging demographics and high turnover rates, resulting in a widening gap in hands-on care (Smith et al., 2021). Due to this shortfall, there is an increased demand for meal assistance technologies that can supplement or replace human caregivers in delivering critical daily care (Jones & Miller, 2023). Successfully implemented, such devices can improve autonomy and overall well-being of disabled individuals.

Innovation

The field of assistive feeding and meal assistance is rather niche, with a limited number of papers published on the topic. Furthermore, the majority of studies focus on serial linkage, commonly known as “articulated”, robotic arms that have joints that allow for rotational motion. The popularity of this type of robotic arm is largely because of its high degrees of freedom (d.o.f.) that the rotating joints permit, allowing for it to be manipulated in many different ways and enabling a diverse range of applications and functionality when it comes to lively tasks.

For our project, we will explore the use of a mobile manipulator type of robot for assistive feeding. A mobile manipulator robot can move linearly along X, Y, and Z coordinates, may have rotation at the ‘wrist’ and can maneuver around its environment on a wheeled base. This is unlike a serial linkage robot which moves by rotating along joints between arm segments. Although the mobile manipulator robots have fewer degrees of freedom than those of serial linkage robots, their mobile base enables a range of access that is sufficient for in-home tasks. Additionally, mobile manipulator robots are specifically designed to be able to operate alongside humans safely. They contain advanced sensors and integrated machine vision technology, enabling the robot to detect and avoid obstacles. These features reduce the risk of accidents, which can lead to more opportunity for combined human-robot achievement, or human-in-the-loop technology. Therefore, it is a safer alternative when compared to traditional robots that require isolated workspaces.

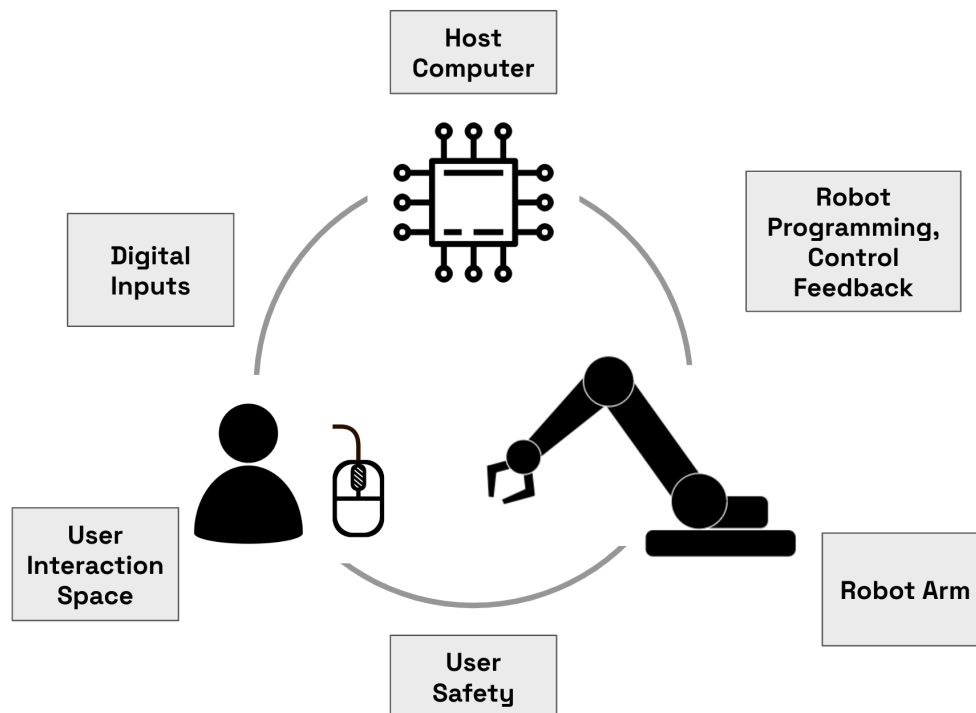
To investigate mobile manipulators, the Stretch by Hello Robot will be used. The robot is a mobile manipulator that is already on the market and in people's homes being used as an assistive device. It is able to maneuver around the home and do certain tasks such as locking the door, picking items off the floor, and delivering food or medication among other objects, and more (Ackerman, 2023). One task that has not been fully developed to market for this particular robot is meal assistance, in which the robot brings bites of food to the user's mouth. We would like to develop this as a feature of the Stretch robot to reduce the need for other assistive devices such as the Kinova, a 7 d.o.f. serial manipulator robotic arm designed to attach to a wheelchair (Campeau-Lecours, 2017). The Stretch implements the task of feeding (Olatunji et al., 2023), but there is no documentation or sources for how the process of coding the robot was done, and thus is not fully developed. Documentation of an auto-feeding system with the stretch

robot can not only increase accessibility for this design but also prove that it can be done with our proposed approach.

In addition to proving that our method works, we will make improvements to its design, specifically automating the food-to-mouth movement. The feeding task that already exists uses manual input controls via directional controls. Most commercial assistive feeding robots use a keyboard or joystick user control system and have the shortest learning curves (Naotunna et al, 2015). However manual controls are not always intuitive, as the robot has more degrees of freedom than the human operator can readily control (Losey et al., 2020). Increasing the autonomy of the feeding can therefore increase the usability of the robot. This process will also be outlined and documented, using Gazebo and other ROS tools to allow for open-source use.

Approach

To develop an autonomous and user-friendly assistive feeding device, we will utilize the Stretch Version 0.2 Robot in the UCLA Biomechatronics lab. This includes a full setup with the robot, integrated computer, monitor, mouse, and necessary software. Our objective is to create a system that autonomously moves the robot from a food station (point A) to the user's mouth (point B) with safety and accuracy, reducing the need for a manual process for every bite. We will use robotics simulations (e.g. Gazebo) for initial testing to ensure safety and functionality, followed by real-world implementation, all through the Robot Operating System (ROS).



The first step of the project will begin with a detailed simulation of the robot's movements in a modeling environment, outlined by the open-source documentation for Stretch simulations. The initial movement path will be designed to simulate a safe, efficient route from point A to point B using built-in reverse kinematics algorithms that optimize the robot's movement without undesired orientations. Testing various kinematics in simulation allows us to identify the most

human-friendly paths to avoid hitting the user or moving outside the designated area. At first, the robot will operate in a controlled starting position on the floor, restricted to specific movements.

Once an optimal path is determined in the simulation, we will implement these movements in the real world. Using ROS, the robot will be calibrated to understand its position relative to point A (food) and point B (mouth). The initial trials will involve preset coordinates to ensure precise movement, minimizing the need for real-time adjustments. For this stage, we will design and prototype a custom end effector to hold various utensils. The calibration will also involve setting movement velocities to a safe threshold, ensuring smooth, controlled motion during feeding.

To accommodate user-specific needs and preferences for utensil positioning, we will add a teleoperated phase after the robot's autonomous movement. Through a joystick interface, the user will be able to fine-tune the utensil's position to achieve an ideal placement relative to their mouth. Detailed online guides on teleoperation and joystick integration for the Stretch robot will be referenced and adapted to our design. To quantify whether this change is necessary or useful, a fully manual mode will be given as well. Both modes will be then compared based on user satisfaction.

If time permits, the next steps will be to include computer vision using the onboard camera for robot adaptability and to identify the food and mouth positions without manual input. The robot's range of movement can also be expanded to allow it to navigate around the room, enabling integration with additional tasks.

Citations

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